

FULL DETAILS (Read-only) -> [Click Here to Create PDF for Current Dataset of Trial](#)

CTRI Number	CTRI/2020/02/023258 [Registered on: 11/02/2020] Trial Registered Prospectively															
Last Modified On:	03/02/2020															
Post Graduate Thesis	No															
Type of Trial	Observational															
Type of Study	Cohort Study															
Study Design	Single Arm Study															
Public Title of Study	Type II Diabetes Mellitus Reversal Study															
Scientific Title of Study	A study on Type II DM reversal by using twins precision nutrition - Retrospective study															
Trial Acronym																
Secondary IDs if Any	<table border="1"> <thead> <tr> <th>Secondary ID</th><th>Identifier</th></tr> </thead> <tbody> <tr> <td>NIL</td><td>NIL</td></tr> </tbody> </table>	Secondary ID	Identifier	NIL	NIL											
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Details of Principal Investigator or overall Trial Coordinator (multi-center study)	<table border="1"> <tr> <td>Name</td><td>DrParamesh Shamanna</td></tr> <tr> <td>Designation</td><td>Director</td></tr> <tr> <td>Affiliation</td><td>Bangalore Diabetes Centre</td></tr> <tr> <td>Address</td><td>Bangalore Diabetes Centre Diabetes & Internal Medicine 426, CMR Rd, HRBR Layout 2nd Block, HRBR Layout, Kalyan Nagar, Bengaluru Bangalore KARNATAKA 560043 India</td></tr> <tr> <td>Phone</td><td>08043479444</td></tr> <tr> <td>Fax</td><td></td></tr> <tr> <td>Email</td><td>dr_paramesh@hotmail.com</td></tr> </table>		Name	DrParamesh Shamanna	Designation	Director	Affiliation	Bangalore Diabetes Centre	Address	Bangalore Diabetes Centre Diabetes & Internal Medicine 426, CMR Rd, HRBR Layout 2nd Block, HRBR Layout, Kalyan Nagar, Bengaluru Bangalore KARNATAKA 560043 India	Phone	08043479444	Fax		Email	dr_paramesh@hotmail.com
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Details of Contact Person Scientific Query	<table border="1"> <tr> <td>Name</td><td>DrArunKumar</td></tr> <tr> <td>Designation</td><td>Professor, Biostatistics</td></tr> <tr> <td>Affiliation</td><td>Bapuji Dental College and Hospital</td></tr> <tr> <td>Address</td><td>Bapuji Dental College and Hospital Dept. of Biostatistics MCC B Block, Davanagere Davanagere KARNATAKA 577004 India</td></tr> <tr> <td>Phone</td><td>08050558443</td></tr> <tr> <td>Fax</td><td></td></tr> <tr> <td>Email</td><td>arun.biostat@gmail.com</td></tr> </table>		Name	DrArunKumar	Designation	Professor, Biostatistics	Affiliation	Bapuji Dental College and Hospital	Address	Bapuji Dental College and Hospital Dept. of Biostatistics MCC B Block, Davanagere Davanagere KARNATAKA 577004 India	Phone	08050558443	Fax		Email	arun.biostat@gmail.com
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Source of Monetary or Material Support	Bangalore Diabetes Centre and Diagnostic Lab, 426, CMR Rd, HRBR Layout 2nd Block, HRBR Layout, Kalyan Nagar, Bengaluru, Karnataka 560043												
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NIL	NIL												
Countries of Recruitment	India												
Sites of Study	<table border="1"> <tr> <th colspan="4">No of Sites = 1</th></tr> <tr> <th>Name of Principal Investigator</th><th>Name of Site</th><th>Site Address</th><th>Phone/Fax/Email</th></tr> <tr> <td>Dr Paramesh S</td><td>Bangalore Diabetes Centre and Diagnostic Lab</td><td>Bangalore Diabetes Centre and Diagnostic Lab Internal Medicine 426, CMR Rd, HRBR Layout 2nd Block, HRBR Layout, Kalyan Nagar, Bengaluru, Karnataka 560043 Bangalore KARNATAKA</td><td>08043479444 dr_paramesh@hotmail.com</td></tr> </table>	No of Sites = 1				Name of Principal Investigator	Name of Site	Site Address	Phone/Fax/Email	Dr Paramesh S	Bangalore Diabetes Centre and Diagnostic Lab	Bangalore Diabetes Centre and Diagnostic Lab Internal Medicine 426, CMR Rd, HRBR Layout 2nd Block, HRBR Layout, Kalyan Nagar, Bengaluru, Karnataka 560043 Bangalore KARNATAKA	08043479444 dr_paramesh@hotmail.com
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Regulatory Clearance Status from DCGI	<table border="1"> <tr> <th>Status</th></tr> <tr> <td>Not Applicable</td></tr> </table>	Status	Not Applicable										
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Health Condition / Problems	<table border="1"> <tr> <th>Health Type</th><th>Condition</th></tr> </table>	Health Type	Condition										
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Studied	Patients	(1) ICD-10 Condition: E118 Type 2 diabetes mellitus with unspecified complications,				
Intervention / Comparator Agent	<table><tr><th>Type</th><th>Name</th><th>Details</th></tr></table>			Type	Name	Details
Type	Name	Details				
Inclusion Criteria	Age From	18.00 Year(s)				
	Age To	80.00 Year(s)				
	Gender	Both				
	Details	1) Own smartphone and capable of using for TPNP 2) Diabetic : Not on DM Rx with HbA1c > 6.4 or on DM Rx with any HbA1c 3) Adequate Hepatic Function : AST/ALT less than or equal 3.0 x ULN 4) Adequate Renal Function define by a. Serum Cretinine less than or equal 1.5 mg/dl b. and/or eGFR > 60 mL/min/1.73 me 5) BMI greater than or equal 18.5 Kg/m2				
ExclusionCriteria	Details	1) Type 1 Diabetes, LADA, MODY, Fibrocalculous pancreatic diabetes (FCPD), Gestational Diabetes Mellitus (GDM) 2) History of keto-acidosis 3) History of myocardial infraction stroke, angina or coronary insufficiency within the last 3 month 4) Need for current chronic corticosteroid therapy (greater than or equal 2.5mg of prednisone per day or an equivalent dose of other anti-inflammatory corticosteroids) 5) Major psychiatric disorder (eg schizophrenia, bipolar disorder) 6) Major surgical procedure or significant traumatic injury within 28 days prior to enrollment date 7) Patients who have undergone any bariatric procedures 8) Pregnant and lactating females				
Method of Generating Random Sequence	Not Applicable					
Method of Concealment	Not Applicable					
Blinding/Masking	Not Applicable					
Primary Outcome	Outcome		TimePoints			
	Diabetes Reversal		Three month			
Secondary Outcome	Outcome		TimePoints			
	To compare HbA1c, HOMA-IR and diabetes medicine between baseline and after 30 and 90 days of twin program Subgroup analysis a.Nutritional Analysis b.Responders and Non responders i.Duration of Type II DM ii.Age iii.HbA1c iv.C-peptide v.DM medication		Three month			

Target Sample Size	Total Sample Size="64" Sample Size from India="64" Final Enrollment numbers achieved (Total)= "Applicable only for Completed/Terminated trials" Final Enrollment numbers achieved (India)= "Applicable only for Completed/Terminated trials"
Phase of Trial	N/A
Date of First Enrollment (India)	15/02/2020
Date of Study Completion (India)	Applicable only for Completed/Terminated trials
Date of First Enrollment (Global)	Date Missing
Date of Study Completion (Global)	Applicable only for Completed/Terminated trials
Estimated Duration of Trial	Years="1" Months="0" Days="0"
Recruitment Status of Trial (Global)	Not Applicable
Recruitment Status of Trial (India)	Not Yet Recruiting
Publication Details	After CTIRI registration, manuscript will be submitted to suitable journal
Individual Participant Data (IPD) Sharing Statement	Will individual participant data (IPD) be shared publicly (including data dictionaries)?
Brief Summary	A study on Type II DM reversal by using twin precision nutrition - Retrospective study Informed consent statement: Patients were not required to give informed consent to the study because the analysis used anonymous clinical data that were obtained. Institutional review board statement: This study was reviewed and approved by the Institutional Review Board of bangalore Diabetes Centre BACKGROUND Type 2 diabetes (T2D) has long been identified as an incurable chronic disease. The best outcome that has been expected is amelioration of diabetes symptoms or slowing its inevitable progression (1). U.K.

Prospective **Diabetes** Study showed improved glycemic control by intensive control compared with conventional therapy during the first year. During the next 5 years, the FPG increased progressively on all therapies. The glycemic deterioration was associated with progressive loss of β -cell function (2). As the disease progresses, it has been shown that glucose-lowering medication use, health care costs, and complications all rise (3).

Approximately 50% of T2D patients will need insulin therapy within ten years of diagnosis (4). Although in the past **diabetes** has been called chronic and irreversible, the paradigm is changing (5,6). The 2016 World Health Organization (WHO) global report on **diabetes** added a section on **diabetes** reversal and acknowledged that it can be achieved through weight loss and calorie restriction (6).

A handful of studies have reported successful weight loss with decreased insulin resistance, plasma glucose, and medication use following a Low-Calorie Diets or Carbohydrate-Restricted Diets (7). The limitation of using a VLCD to manage a chronic disease is that this type of diet is necessarily temporary, given that it provides less than 800 kcal $\cdot$ day $^{-1}$ and thus is unsustainable in the long term (8). Studies have suggested that physiological and metabolic adaptation of the body in response to caloric restriction may shift energy balance and hormonal regulation of weight toward weight regain after weight loss (9). Studies utilizing carbohydrate restriction observed improved glycemic control and cardiometabolic markers, but were often short-term trials of small groups, excluded subjects prescribed insulin, or infrequently monitored or achieved ketosis (10–14).

Internet of Things (IoT) can be viewed as a dynamic network where physical and virtual objects are interconnected together. Health monitoring systems based on Internet-of-things (IoT) have been recently introduced to improve the quality of health care services (15).

Despite their importance, no method exists for predicting PPGRs to food. The current practice is to use the meal carbohydrate content, even though it is a poor predictor of the PPGR (16). Other methods aimed at estimating PPGRs are the glycemic index (17). However, the few small-scale ($n = 23$ – 40) studies that examined interpersonal differences in PPGRs found high variability in the response of different people to the same food (18).

This is the first study looking at **diabetes** reversal while patients are on AGP. We asked whether personally tailored dietary interventions based on our algorithm could improve PPGR.

OBJECTIVES

Phase I

1. To compare HbA1c, HOMA-IR and **diabetes** medicine between baseline and after 30 and 90 days of twin program

Subgroup analysis

- a. Nutritional Analysis
- b. Responders and Non responders

- i. Duration of Type II DM
 - ii. Age
 - iii. HbA1c
 - iv. C-peptide
 - v. DM medication
- 2. To compare weight, BMI and vitals before and after twin program
- 3. To compare cardiovascular marker before and after twin program
 - a. hs-CRP
 - b. Apo-A1
 - c. Apo-B
 - d. Apo-B/A1
 - e. Homocysteine
 - f. ESR
 - g. WBC
 - h. Ferritin
 - i. Total Cholesterol
 - j. LDL
 - k. HDL
 - l. Triglyceride
 - m. TG/HDL
- 4. To compare BP, MAP, PP and HT medication before and after twin program
- 5. To compare HbA1c, **Diabetes** medications, BP, Anti-Hypertensive medications, weight, BMI, Cardiovascular markers before and after twin program

Phase II

- 6. To study Inter & Intra subject glycemic variability on identical meals
- 7. To find concordance between the predicted 1dag and actual 1dag
- 8. To find ketone levels as a prognostic marker for outcomes
- 9. To compare LFT before and after twin program
- 10. To compare glycemic response to red foods before and after twin program
- 11. To compare the glycemic variability before and after twin program
- 12. To find relation between Calorie, macronutrients and outcome
- 13. To compare the sleep pattern before and after twin program
 - a. Sleep and outcome
- 14. To compare the biochemical parameters before and after twin program
- 15. To find relationship between physical activities and AGP

MATERIAL AND METHODS

Study Design

Retrospective Study

Patients diagnosed with Type II DM and part of twin Precision Nutrition (TPN) Program and have at least three months of followup will be considered for this study.

Inclusion Criteria

1. Age between 18 and 80 of both sex
2. Own smartphone and capable of using for TPNP
3. Diabetic: Not on DM Rx with HbA1c > 6.4 or on DM Rx with any HbA1c
4. Adequate Hepatic Function : AST/ALT $\leq$ 3.0 x ULN
5. Adequate Renal Function defined by
 - a. Serum Creatinine $\leq$ 1.5 mg/dl
 - b. and/or eGFR > 60 mL/min/1.73 m²
6. BMI: $\leq$ 18.5 Kg/m²

Exclusion Criteria

1. Type 1 Diabetes, LADA, MODY, FCPD [Fibrocalculous pancreatic diabetes], GDM[Gestational Diabetes Mellitus]
2. History of keto-acidosis
3. History of myocardial infarction, stroke, angina or coronary insufficiency within the last 3 month
4. Need for current chronic corticosteroid therapy ($\leq$ 2.5 mg of prednisone per day or an equivalent dose of other anti-inflammatory corticosteroids)
5. Major psychiatric disorder (e.g., schizophrenia, bipolar disorder)
6. Major surgical procedure or significant traumatic injury within 28 days prior to Enrollment Date
7. Patients who have undergone any Bariatric procedures
8. Pregnant and lactating females

TPN Program

twin precision treatment use the latest technology and wearable sensors to constantly track and analyze body's health signals to personalize participants treatment. The twin platform converges the latest advancements in Internet of Things (IoT) and Artificial Intelligence (AI) to analyze body sensor and blood work data. This technology-enabled medical service is called TPN Program.

Patients who enrolled in TPN Program according to the eligibility criteria will be analysed retrospectively. TPN Program was an Outpatients program. In the first enrollment visit of the TPN Program, history, vitals, ECG and Biothesiometry was performed. Fasting blood draws done at baseline, 30 days, 90 days, 159 days, 251 days and 342 days respectively. The baseline diet was recorded by recall. Patients performed daily Ketone testing with Keto-Mojo Bluetooth Meter Kit. Patients were encouraged to maintain BOHB readings $\leq$ 0.5 mmol/L. Patients were asked to wear fitbit charge 2 wristband continuously to record Sleep parameters, heart rate, step count and other fitness parameters. Patients were asked to record the blood pressure daily using digital bluetooth enabled bp recording TAIDOC TD-

3140. Patients measured their weight daily morning after the first void of urine using Powermax BCA-130 Bluetooth Smart Scale. Continuous glucose monitoring (Ambulatory Glucose Profile, AGP) was performed throughout the study using Abbott Libre Pro CGM Diabetes Sensor which was integrated into the TPNP web based software. All these data was transmitted through cellular network to the software. This information was made available to the patients as biometric feedback via the twin app. Patients were provided access to the software via the twin app. They were asked to record the food intake on the app. The Artificial Intelligence (A.I) analysed the macro and micronutrients from the database. Using Machine Learning (M.L), a subtype of artificial intelligence, the billions of data points were analyzed to see what drove the glucose response to specific foods for each participant. Factors found to be involved in glycemic response were analysed for each participant. Participants were then provided with a set of specific food recommendations in order to avoid glucose spikes. Using continuous glucose monitors (CGM) data, blood glucose values of the participants were predicted using machine learning algorithms and data fusion techniques. Each food item within every meal was logged along with its weight by selecting it from a database of more than 2000 foods with full nutritional values based on the usda database that we further improved and expanded with additional items from certified sources. Machine Learning (M.L) algorithm devised to integrate these multi-dimensional data and accurately predicts personalized PPGR (Postprandial Glucose Response). Dietary intake is a central determinant of blood glucose levels, and thus, in order to achieve normal glucose levels it is imperative to make food choices that induce normal postprandial (postmeal) glycemic responses. Thus the platform was able to suggest the right food to the right patient while connected to the CGM.

The twin program aimed to provide the right macronutrients and micronutrients. Net Carbs were restricted to 47 grams per day and avoid sugar intake. Patients were encouraged to consume protein above 1 gram per kg of lean body weight. Patients were encouraged to consume saturated fats and avoid Trans Fats and vegetable oils. There was no cap on calorie consumption and patients were allowed to consume food ad-libitum to satiety keeping in mind the limit on carbs. Patients were also provided kefir and pickles to improve the gut bacteria. Patients were provided supplements to provide Magnesium, other Vitamins and Minerals. Nutritional counseling was provided by trained coaches through the app and via telephone.

Patients insulin dose was reduced at the beginning. All Oral Hypoglycemic agents except Metformin and DPP4 inhibitors were stopped. Further medication management was based on customized guidelines by the Physician. All patients symptoms were recorded and analysed by the Physicians.

Statistical Analysis

Collected data will be analysed using R-software version 3.6.1. Normally distributed continuous variables will be presented as mean and standard deviation and for non-normal continuous variables will be presented as median and range. Categorical variables will be presented as count and per cent. To compare mean between baseline and 90 days, paired t-test will be used. To compare means between more than two time points, one-way repeated measure ANOVA will be used. For non-normal continuous variables Wilcoxon signed-rank test and Friedman test will be used respectively. $p < 0.05$ will be considered statistically significant.

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